hERG Inhibition Prediction

Feature-Diverse Ensemble with Backward Elimination

Public LB 8.77171

5-Fold OOF · Multi-start Nelder-Mead · Top-5 Ensemble

Overall Strategy

① Build Diverse Models

Combine different feature types (descriptor, fingerprint, MACCS, ChemBERTa)
and different model types (MLP, LightGBM) to cover orthogonal information spaces.

② Maximize Ensemble Size

Train 10 models and run Multi-start Nelder-Mead weight optimization on OOF predictions.
More models → richer hypothesis space.

③ Backward Elimination

Identify near-zero weight models from the 10-model ensemble.
Remove them step-by-step and validate on Public LB → prevents OOF overfitting.

④ Optimal Subset = Top-5

Top-5 model ensemble achieved the best Public LB: 8.8629921078
Fewer models → better generalization to private test distribution.

Diverse Feature × Model Combinations

PCGrad Intuition: Only add models with orthogonal predictions → no redundancy, maximum information gain

Descriptor MLP

Feature

RDKit Descriptor (210-dim)

Architecture

MLP 256→128→64→1

OOF MAE

8.903

Desc+FP+MACCS MLP

Feature

Descriptor+FP+MACCS (2,424-dim)

Architecture

MLP 1024→512→256→1

OOF MAE

9.193

Desc+FP LightGBM

Feature

Descriptor+Fingerprint (2,258-dim)

Architecture

LightGBM

OOF MAE

9.081

Fingerprint LGB FS500

Feature

Morgan FP → Top-500 Feature Select

Architecture

LightGBM

OOF MAE

9.303

ChemBERTa MLP

Feature

ChemBERTa Embedding (768-dim)

Architecture

MLP 1024→512→256→1

OOF MAE

9.433

Ensemble: Multi-start Nelder-Mead

Weight Optimization

① OOF predictions stacked: (N, 10) matrix

② Objective: minimize MAE(y_true, W·OOF)

③ 10 starting points:

- Uniform weights (1/n, …)
- Inverse-MAE weights
- 8 × random Dirichlet

④ Best result across all starts selected

⑤ W normalized: |w| / Σ|w| (sum = 1)

10-Model Optimal Weights

Backward Elimination: Finding the Optimal Subset

Remove near-zero weight models → re-optimize → compare Public LB

Ensemble	# Models	OOF MAE	Public LB	vs. 9-model Best
9-model (baseline)	9	8.598169	8.9172977293	—
10-model	10	8.530895	—	submitted
top7	7	8.530890	—	≈ 10model
top5 ★ BEST	**5**	**8.533049**	**8.77171**	**▲ +0.054**
top4	4	8.546300	worse	▼ degraded
top3	3	8.563052	worse	▼ degraded

Key Insight: OOF MAE slightly worse for top5, but Public LB improved → Nelder-Mead overfits with too many models

Final Solution Architecture

RDKit Descriptor	Morgan FP →FS500	Desc+FP +MACCS	Desc+FP	ChemBERTa Embed
MLP 256→128→64	LightGBM FS500	MLP 1024→512→256	LightGBM	MLP 1024→512→256

5-Fold OOF predictions

Multi-start Nelder-Mead Weight Optimization (10 starts)

w = argmin MAE(y_true, OOF_matrix @ w)

Backward Elimination (10 → 7 → 5 → 4)

Remove weight≈0 models · validate on PL

Top-5 Ensemble | PL: 8.77171

Conclusion

01 Diversity First

Combined 5 different feature spaces × 2 model types.
PCGrad intuition: orthogonal predictions → no redundancy.

02 Nelder-Mead as Selector

Multi-start optimization naturally assigns weight ≈ 0
to redundant/weak models — a built-in feature selector.

03 Backward Elimination = Regularization

Removing weight≈0 models reduced OOF overfitting.
Top-5 beat Top-10 on Public LB by +0.054 MAE.

04 Key Takeaway

In ensemble learning, more models ≠ better generalization.
Optimal subset size must be validated on held-out data.

Final Public LB: 8.77171